AMA3020 Project: Supplementary Derivations

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This document contains the full algebraic derivation of the recurrence relation and equilibrium value for the unequal-volume case discussed in the main report.
Let the milk jug have volume V_1 and the coffee jug have volume V_2, with $V_1 \neq V_2$. At each round, a fixed volume a is transferred from the milk jug to the coffee jug, mixed thoroughly, and the same volume a is transferred back.

Let x_n denote the amount of coffee in the milk jug after n rounds.

Step 1: Coffee remaining in the milk jug after the first transfer

After the initial transfer, the milk jug contains x_n coffee in total volume V_1, so the coffee fraction is

$$\frac{x_n}{V_1}.$$

When volume a is removed, the amount of coffee removed is

$$a \cdot \frac{x_n}{V_1}.$$

Hence the coffee remaining in the milk jug is

$$x_n - a\frac{x_n}{V_1} = x_n \left(1 - \frac{a}{V_1}\right).$$

Step 2: Coffee in the coffee jug after receiving a

Initially the coffee jug contains $V_2 - x_n$ coffee (since total coffee is conserved).
The coffee transferred into it from the milk jug is

$$a\frac{x_n}{V_1}.$$

Thus the total coffee in the coffee jug after receiving a is

$$(V_2 - x_n) + a\frac{x_n}{V_1}.$$

The total volume of the coffee jug is now $V_2 + a$, so the coffee fraction is

$$\frac{(V_2 - x_n) + a\frac{x_n}{V_1}}{V_2 + a}.$$

Step 3: Coffee returned to the milk jug

When volume a is transferred back, the amount of coffee returned is

$$a \cdot \frac{(V_2 - x_n) + a\frac{x_n}{V_1}}{V_2 + a}.$$

Step 4: Recurrence relation

The total coffee in the milk jug after the round is therefore

$$x_{n+1} = x_n \left(1 - \frac{a}{V_1}\right) + a \cdot \frac{(V_2 - x_n) + a\frac{x_n}{V_1}}{V_2 + a}.$$

Expand the second fraction:

$$x_{n+1} = x_n \left(1 - \frac{a}{V_1}\right) + \frac{aV_2}{V_2 + a} - \frac{ax_n}{V_2 + a} + \frac{a^2 x_n}{V_1(V_2 + a)}.$$

Collect terms involving x_n:

$$x_{n+1} = x_n \left[\left(1 - \frac{a}{V_1}\right) - \frac{a}{V_2 + a} + \frac{a^2}{V_1(V_2 + a)}\right] + \frac{aV_2}{V_2 + a}.$$

Factorise the last two terms in the bracket:

$$-\frac{a}{V_2 + a} + \frac{a^2}{V_1(V_2 + a)} = -\frac{a}{V_2 + a}\left(1 - \frac{a}{V_1}\right).$$

Substituting this back gives

$$x_{n+1} = x_n \left[\left(1 - \frac{a}{V_1}\right) - \frac{a}{V_2 + a}\left(1 - \frac{a}{V_1}\right)\right] + \frac{aV_2}{V_2 + a}.$$

Now factor out $\left(1 - \frac{a}{V_1}\right)$:

$$x_{n+1} = x_n \left(1 - \frac{a}{V_1}\right)\left(1 - \frac{a}{V_2 + a}\right) + \frac{aV_2}{V_2 + a}.$$

Since

$$1 - \frac{a}{V_2 + a} = \frac{V_2}{V_2 + a},$$

we obtain

$$x_{n+1} = \frac{V_2}{V_2 + a}\left(1 - \frac{a}{V_1}\right) x_n + \frac{aV_2}{V_2 + a}.$$

Thus the recurrence has the form

$$x_{n+1} = Ax_n + B,$$

where

$$A = \frac{V_2}{V_2 + a}\left(1 - \frac{a}{V_1}\right), \qquad B = \frac{aV_2}{V_2 + a}.$$

Equilibrium

At equilibrium, $x_{n+1} = x_n = x_\infty$, so

$$x_\infty = Ax_\infty + B.$$

Rearranging gives

$$x_\infty(1-A) = B, \qquad x_\infty = \frac{B}{1-A}.$$

Compute $1 - A$:

$$\begin{aligned}
1 - A &= 1 - \frac{V_2}{V_2 + a}\left(1 - \frac{a}{V_1}\right) \\
&= \frac{(V_2 + a) - V_2 + \frac{aV_2}{V_1}}{V_2 + a} \\
&= \frac{a\left(1 + \frac{V_2}{V_1}\right)}{V_2 + a} \\
&= \frac{a(V_1 + V_2)}{V_1(V_2 + a)}.
\end{aligned}$$

Substituting into $x_\infty = \frac{B}{1-A}$:

$$\begin{aligned}
x_\infty &= \frac{\frac{aV_2}{V_2+a}}{\frac{a(V_1+V_2)}{V_1(V_2+a)}} \\
&= \frac{aV_2}{V_2 + a} \cdot \frac{V_1(V_2 + a)}{a(V_1 + V_2)} \\
&= \frac{V_1 V_2}{V_1 + V_2}.
\end{aligned}$$

Hence the equilibrium proportion of coffee in the milk jug is

$$p_\infty = \frac{x_\infty}{V_1} = \frac{V_2}{V_1 + V_2}.$$